OVERFLOW RULE

```
// If any value or sum/product can exceed ~2×10^9, use long long
```

FAST I/O

```
#define SpeedxKH ios_base::sync_with_stdio(false); cin.tie(NULL);
```

2D VECTOR TRAVERSE

```
for (int i = 0; i < n; i++) for (int j = 0; j < m; j++) cout << g[i][j] << " ";
```

VECTOR – DECLARATION

```
vector<int> v;  // empty vector
vector<int> v(n);  // size n, garbage    vector<int> v(n, 0);  // size n, all 0
vector<vector<int>> g(n, vector<int>(m, 0));  // 2D vector (n x m)
```

VECTOR – INPUT / OUTPUT

```
for (int i=0;i<n;i++) cin>>v[i];         for (int x:v) cout<<x<<" ";
```

VECTOR – MODIFY

```
v.push_back(x);  // add to end           v.pop_back();  // remove last
v.insert(v.begin()+pos, x);  //insert at pos v.erase(v.begin()+pos);  //erase at pos
v.clear();  // remove all                v.resize(newSize);  // resize vector
v.size();  // element count              v.empty();  // true if size==0
v.front();  // first element             v.back();  // last element
```

VECTOR – MAX / MIN / SUM

```
*max_element(v.begin(),v.end());            //max value
*min_element(v.begin(),v.end());            //min value
max_element(v.begin(),v.end())-v.begin();  // index of max
min_element(v.begin(),v.end())-v.begin();  // index of min
accumulate(v.begin(), v.end(), 0LL);         // sum, avoid overflow
```

VECTOR – SORT / REVERSE / SEARCH

```
sort(v.begin(), v.end());  // ascending
sort(v.begin(), v.end(), greater<int>());  // descending
reverse(v.begin(),v.end());  // reverse order
count(v.begin(),v.end(),x);  // count of x
find(v.begin(), v.end(), x);  // iterator (==v.end() if not found)
```

VECTOR – UNIQUE (remove duplicates, needs sorted)

```
sort(v.begin(),v.end());                 v.erase(unique(v.begin(),v.end()),v.end());
```

SORT INDICES BY VALUE

```
vector<int> idx(n); iota(idx.begin(),idx.end(),0);
sort(idx.begin(),idx.end(),[&](int i,int j){return a[i]<a[j];});
```

EXTRA VECTOR ESSENTIALS

```
v1.insert(v1.end(),v2.begin(),v2.end());  // merge v2 into v1
swap(v1, v2);  // O(1) swap
vector<int> v = {1, 2, 3};  // direct init
vector<int> v(v2.begin(), v2.end());  // copy from another vector
find(v.begin(), v.end(), x) != v.end();  // true if x exists
v.erase(remove(v.begin(), v.end(), x), v.end());  // remove all occurrences of x
fill(v.begin(), v.end(), x);  // set all elements to x

#define ll long long  #define ull unsigned long long #define ld long double
#define endl '\n'
const int MOD = 1e9 + 7;
const int INF = 1e9;
const ll LLINF = 1e18;
```